Table 1. Main results on Qwen3-VL. Each entry reports accuracy (%) followed by speedup (×) in parentheses. Four model sizes, six benchmarks, and eight methods yield 192 cells.

Model	Benchmark	Native	Ours	F3A	VisionZip	DivPrune	ReplaceMe	FastV	S2Prune
2B	MMStar	55.53 (1.00)	**52.33 (6.99)**	29.53 (2.59)	30.80 (6.26)	30.73 (6.29)	42.67 (7.03)	27.73 (6.22)	31.20 (5.37)
2B	AI2D	76.00 (1.00)	**74.19 (5.71)**	59.39 (2.65)	59.33 (5.27)	59.13 (5.29)	51.10 (5.71)	58.91 (5.25)	60.27 (4.72)
2B	MMBench-EN	77.47 (1.00)	**75.51 (6.31)**	25.51 (2.22)	26.79 (5.61)	32.14 (5.62)	16.58 (6.35)	18.62 (5.56)	32.82 (4.76)
2B	CV-Bench	79.50 (1.00)	**78.69 (7.34)**	44.52 (3.55)	42.96 (6.85)	42.99 (6.88)	64.16 (7.36)	39.25 (6.82)	46.04 (6.16)
2B	VisualPuzzles	31.25 (1.00)	**30.48 (5.80)**	30.74 (2.66)	28.68 (5.32)	29.28 (5.34)	25.34 (5.77)	29.37 (5.29)	30.05 (4.76)
2B	VisuLogic	26.40 (1.00)	**26.60 (4.83)**	25.10 (2.12)	23.80 (4.37)	25.00 (4.38)	23.50 (4.78)	25.50 (4.34)	24.00 (3.87)
4B	MMStar	62.53 (1.00)	**61.33 (6.19)**	32.53 (2.72)	33.53 (5.95)	32.93 (5.91)	58.53 (6.37)	32.87 (5.89)	32.40 (5.19)
4B	AI2D	82.29 (1.00)	**79.76 (5.41)**	63.21 (2.79)	62.63 (5.25)	63.05 (5.22)	79.44 (5.55)	62.50 (5.21)	64.15 (4.73)
4B	MMBench-EN	83.93 (1.00)	**82.14 (5.51)**	38.18 (2.33)	39.37 (5.29)	43.71 (5.29)	81.63 (5.75)	36.65 (5.28)	46.60 (4.58)
4B	CV-Bench	85.05 (1.00)	**83.13 (7.40)**	48.86 (3.89)	47.83 (7.19)	48.97 (7.23)	82.33 (7.65)	50.23 (7.22)	50.40 (6.50)
4B	VisualPuzzles	31.42 (1.00)	**32.45 (5.40)**	31.25 (2.78)	30.65 (5.23)	31.51 (5.23)	30.82 (5.53)	30.22 (5.19)	31.08 (4.71)
4B	VisuLogic	25.40 (1.00)	**27.60 (4.32)**	24.80 (2.19)	24.90 (4.18)	25.10 (4.17)	26.40 (4.42)	25.60 (4.14)	25.00 (3.76)
8B	MMStar	65.00 (1.00)	**62.73 (5.46)**	32.00 (2.69)	31.60 (5.37)	32.40 (5.37)	62.33 (5.59)	30.73 (5.27)	33.67 (4.82)
8B	AI2D	84.59 (1.00)	**82.32 (5.14)**	64.93 (2.86)	65.61 (5.04)	65.09 (5.04)	82.64 (5.23)	64.73 (4.99)	64.93 (4.62)
8B	MMBench-EN	84.86 (1.00)	**82.74 (4.59)**	36.99 (2.22)	35.12 (4.53)	40.22 (4.51)	83.16 (4.72)	30.44 (4.46)	45.24 (4.03)
8B	CV-Bench	86.00 (1.00)	**83.14 (7.27)**	54.38 (4.11)	54.21 (7.17)	53.16 (7.20)	83.22 (7.40)	52.06 (7.13)	58.44 (6.61)
8B	VisualPuzzles	34.33 (1.00)	**34.50 (5.06)**	29.20 (2.81)	28.68 (5.01)	28.00 (5.00)	34.42 (5.16)	29.02 (4.95)	28.25 (4.60)
8B	VisuLogic	25.40 (1.00)	**25.50 (3.79)**	28.00 (2.09)	26.10 (3.77)	27.10 (3.76)	26.10 (3.87)	26.40 (3.71)	28.80 (3.46)
32B	MMStar	73.13 (1.00)	**70.00 (4.27)**	33.67 (3.14)	68.27 (4.32)	53.40 (4.40)	68.20 (4.48)	65.00 (4.29)	48.20 (4.32)
32B	AI2D	88.15 (1.00)	**85.78 (4.75)**	66.87 (3.57)	85.14 (4.82)	73.12 (4.86)	85.82 (4.96)	82.84 (4.79)	67.39 (4.77)
32B	MMBench-EN	88.35 (1.00)	**88.01 (3.43)**	43.28 (2.51)	86.39 (3.47)	81.55 (3.55)	87.50 (3.60)	86.31 (3.45)	76.02 (3.47)
32B	CV-Bench	88.88 (1.00)	**85.84 (7.16)**	53.83 (5.44)	85.95 (7.28)	75.42 (7.45)	85.71 (7.48)	84.80 (7.22)	69.41 (7.19)
32B	VisualPuzzles	40.84 (1.00)	**38.18 (4.83)**	32.53 (3.67)	36.90 (4.83)	31.51 (4.97)	38.18 (5.04)	33.73 (4.81)	33.39 (4.88)
32B	VisuLogic	26.30 (1.00)	**28.40 (3.30)**	23.60 (2.50)	27.30 (3.28)	26.30 (3.37)	27.30 (3.45)	27.10 (3.26)	23.70 (3.33)

Bold identifies Ours; it does not denote the best score. Native is the baseline for each model and benchmark. All cells cover the complete input set with three repetitions (1,322,016 calls in total). Speedup is native latency divided by method latency; it excludes loading, packing, and initial graph capture. MMBench-EN uses provided-pass group accuracy, not full official CircularEval; 53 groups are incomplete. CV-Bench uses source-balanced accuracy; the other benchmarks use forced-choice exact accuracy. Three repetitions are timing replicates, not independent training seeds.

Table 2. End-to-end steady-state latency on Qwen3-VL. Values are in ms.

Model	Benchmark	Native	Ours	F3A	VisionZip	DivPrune	ReplaceMe	FastV	S2Prune
2B	MMStar	43.074	6.161	16.607	6.876	6.853	6.125	6.927	8.018
2B	AI2D	51.414	9.006	19.366	9.752	9.715	8.998	9.801	10.898
2B	MMBench-EN	35.965	5.696	16.186	6.408	6.399	5.662	6.473	7.562
2B	CV-Bench	69.470	9.466	19.567	10.147	10.100	9.433	10.179	11.271
2B	VisualPuzzles	51.815	8.929	19.468	9.744	9.708	8.976	9.788	10.889
2B	VisuLogic	39.458	8.164	18.644	9.033	9.011	8.261	9.087	10.190
4B	MMStar	48.980	7.917	17.998	8.232	8.292	7.691	8.317	9.441
4B	AI2D	58.256	10.768	20.850	11.098	11.152	10.487	11.177	12.317
4B	MMBench-EN	41.143	7.465	17.657	7.780	7.781	7.154	7.798	8.975
4B	CV-Bench	82.260	11.114	21.121	11.434	11.379	10.752	11.394	12.651
4B	VisualPuzzles	58.482	10.832	21.019	11.190	11.190	10.582	11.266	12.405
4B	VisuLogic	45.110	10.443	20.596	10.801	10.816	10.205	10.899	11.999
8B	MMStar	53.471	9.788	19.842	9.950	9.964	9.557	10.138	11.103
8B	AI2D	64.474	12.552	22.573	12.801	12.802	12.336	12.914	13.959
8B	MMBench-EN	43.129	9.396	19.461	9.519	9.567	9.135	9.680	10.690
8B	CV-Bench	94.549	13.006	22.977	13.184	13.133	12.784	13.256	14.308
8B	VisualPuzzles	64.515	12.744	22.988	12.878	12.904	12.497	13.042	14.026
8B	VisuLogic	47.067	12.407	22.500	12.497	12.532	12.168	12.698	13.618
32B	MMStar	95.159	22.287	30.289	22.045	21.644	21.240	22.173	22.017
32B	AI2D	116.910	24.594	32.758	24.274	24.073	23.591	24.428	24.515
32B	MMBench-EN	75.616	22.075	30.069	21.786	21.323	21.021	21.923	21.766
32B	CV-Bench	180.651	25.216	33.179	24.830	24.252	24.151	25.004	25.108
32B	VisualPuzzles	122.820	25.446	33.452	25.407	24.716	24.371	25.518	25.154
32B	VisuLogic	83.441	25.256	33.321	25.439	24.750	24.184	25.571	25.048

Timing includes image I/O, preprocessing, model execution, CPU output, and JSON serialization, averaged over repetitions 2 and 3. Methods execute sequentially on the same GPU and model process; Native is freshly loaded after the accelerated methods. MMBench-EN uses provided-pass group accuracy, not full official CircularEval; 53 groups are incomplete. CV-Bench uses source-balanced accuracy; the other benchmarks use forced-choice exact accuracy. Three repetitions are timing replicates, not independent training seeds.

Table 3. Accuracy loss relative to Native on Qwen3-VL. Values are in percentage points.

Model	Benchmark	Native	Ours	F3A	VisionZip	DivPrune	ReplaceMe	FastV	S2Prune
2B	MMStar	0.000	3.200	26.000	24.733	24.800	12.867	27.800	24.333
2B	AI2D	0.000	1.813	16.613	16.677	16.872	24.903	17.098	15.738
2B	MMBench-EN	0.000	1.956	51.956	50.680	45.323	60.884	58.844	44.643
2B	CV-Bench	0.000	0.807	34.980	36.539	36.515	15.341	40.256	33.465
2B	VisualPuzzles	0.000	0.771	0.514	2.568	1.969	5.908	1.884	1.199
2B	VisuLogic	0.000	-0.200	1.300	2.600	1.400	2.900	0.900	2.400
4B	MMStar	0.000	1.200	30.000	29.000	29.600	4.000	29.667	30.133
4B	AI2D	0.000	2.526	19.074	19.657	19.236	2.850	19.786	18.135
4B	MMBench-EN	0.000	1.786	45.748	44.558	40.221	2.296	47.279	37.330
4B	CV-Bench	0.000	1.924	36.194	37.222	36.084	2.729	34.822	34.651
4B	VisualPuzzles	0.000	-1.027	0.171	0.771	-0.086	0.599	1.199	0.342
4B	VisuLogic	0.000	-2.200	0.600	0.500	0.300	-1.000	-0.200	0.400
8B	MMStar	0.000	2.267	33.000	33.400	32.600	2.667	34.267	31.333
8B	AI2D	0.000	2.267	19.657	18.977	19.495	1.943	19.851	19.657
8B	MMBench-EN	0.000	2.126	47.874	49.745	44.643	1.701	54.422	39.626
8B	CV-Bench	0.000	2.859	31.621	31.784	32.836	2.780	33.935	27.555
8B	VisualPuzzles	0.000	-0.171	5.137	5.651	6.336	-0.086	5.308	6.079
8B	VisuLogic	0.000	-0.100	-2.600	-0.700	-1.700	-0.700	-1.000	-3.400
32B	MMStar	0.000	3.133	39.467	4.867	19.733	4.933	8.133	24.933
32B	AI2D	0.000	2.364	21.276	3.012	15.026	2.332	5.311	20.758
32B	MMBench-EN	0.000	0.340	45.068	1.956	6.803	0.850	2.041	12.330
32B	CV-Bench	0.000	3.039	35.057	2.937	13.458	3.172	4.086	19.473
32B	VisualPuzzles	0.000	2.654	8.305	3.938	9.332	2.654	7.106	7.449
32B	VisuLogic	0.000	-2.100	2.700	-1.000	0.000	-1.000	-0.800	2.600

Loss = Native accuracy − method accuracy; negative values indicate an accuracy increase. Accuracy equality does not imply identical per-example predictions. MMBench-EN uses provided-pass group accuracy, not full official CircularEval; 53 groups are incomplete. CV-Bench uses source-balanced accuracy; the other benchmarks use forced-choice exact accuracy. Three repetitions are timing replicates, not independent training seeds.

Table 4. Pooled Qwen3-VL speedups. Ratios use sample-weighted latency over all six benchmarks.

Model	Native	Ours	F3A	VisionZip	DivPrune	ReplaceMe	FastV	S2Prune
2B	1.000	6.291	2.672	5.741	5.760	6.301	5.707	5.053
4B	1.000	5.866	2.834	5.671	5.665	6.055	5.648	5.047
8B	1.000	5.378	2.851	5.301	5.296	5.493	5.238	4.819
32B	1.000	4.697	3.511	4.747	4.841	4.913	4.719	4.738

These measurements belong to the main comparison cohort. Earlier standalone measurements are retained as historical results and are not substituted into this table.

Table 5. Frozen comparison budgets. Calibration selects the available candidate closest to the latency of Ours using 32 image-disjoint GQA examples without labels.

Model	Method	Selected budget	Latency / Ours	Within $\pm 10\%$
2B	F3A	2% retained	2.428	No
2B	VisionZip	2% retained	1.082	Yes
2B	DivPrune	2% retained	1.081	Yes
2B	ReplaceMe	2 layers removed	0.977	Yes
2B	FastV	2% retained	1.091	Yes
2B	S2Prune	2% retained	1.249	No
4B	F3A	2% retained	2.202	No
4B	VisionZip	2% retained	1.036	Yes
4B	DivPrune	2% retained	1.038	Yes
4B	ReplaceMe	2 layers removed	0.976	Yes
4B	FastV	5% retained	1.044	Yes
4B	S2Prune	2% retained	1.182	No
8B	F3A	2% retained	1.990	No
8B	VisionZip	2% retained	1.023	Yes
8B	DivPrune	2% retained	1.026	Yes
8B	ReplaceMe	2 layers removed	0.975	Yes
8B	FastV	2% retained	1.036	Yes
8B	S2Prune	2% retained	1.133	No
32B	F3A	2% retained	1.362	No
32B	VisionZip	70% retained	0.988	Yes
32B	DivPrune	20% retained	0.983	Yes
32B	ReplaceMe	4 layers removed	0.954	Yes
32B	FastV	70% retained	0.996	Yes
32B	S2Prune	10% retained	1.000	Yes

All competitors are Qwen3-VL adaptations on a shared accelerated backend, not measurements of the authors' original end-to-end systems. Several 2B–8B settings retain only 2% of visual tokens. F3A and some S2Prune settings remain unmatched; severe losses cannot establish that the original methods are intrinsically ineffective. Accelerated paths include frozen input/prompt approximations and FP8 (precise accumulation for 8B/32B); Native uses native inputs and BF16. These are not lossless, same-input kernel-only comparisons.

Table 6. Supplementary cross-family transfer and adaptation studies. Qwen3-VL remains the main evaluation; other families provide supporting evidence.

Family	Model	Speedup (×)	Maximum loss (pp)	Measurement scope
Ministral3	3B	3.2318	4.3104	Request-stream speedup
Ministral3	8B	3.5799	4.1909	Request-stream speedup
Ministral3	14B	3.7294	4.4483	Request-stream speedup
Gemma3	4B	3.255639	0.000000	B2 request-stream throughput
Gemma3	12B	2.555902	0.170068	B2 request-stream throughput
Gemma3	27B	1.987976	0.000000	B2 request-stream throughput
InternVL3.5	8B	1.64–1.82	Approx. 4.34	Per-benchmark warm latency
LLaVA-1.5	7B	3.38	0 on GQA64	Pilot only; no six-benchmark run

These studies are not controlled single-component ablations or a unified speed ranking. Ministral includes request-time capture; Gemma compares B2 throughput against serial HF; InternVL reuses a historical fixed-image baseline; Qwen excludes initial capture. Ratios must not be averaged across families. The 42 full-benchmark comparisons exclude the LLaVA pilot.

Table 7. InternVL3.5-8B adaptation. Accuracy is in percent; improvement is relative to the earlier transfer configuration.

Benchmark	Native	Earlier	Adapted	Improvement (pp)	Latency (ms)	Speedup (×)
MMStar	64.27	48.33	59.93	+11.60	20.71	1.80
AI2D	80.63	70.34	79.89	+9.55	25.07	1.66
MMBench-EN	80.61	64.97	78.32	+13.35	20.07	1.82
CV-Bench	81.11	69.04	80.03	+10.98	25.39	1.66
VisualPuzzles	30.57	29.28	29.97	+0.68	25.37	1.64
VisuLogic	29.50	23.80	27.30	+3.50	21.83	1.73

The adapted model uses 448-pixel inputs, 26/36 language layers, all vision layers, calibrated static quantization, and a rank-32 recovery module. Calibration, resolution, and teacher/retraining change jointly; retained layer indices are unchanged. Latency is compared with historical H100 fixed-448 single-image controls, not a contemporaneous paired rerun or default dynamic tiling.

Table 8. Ministral3 supplementary full-benchmark results. Loss is baseline accuracy minus method accuracy (percentage points).

Model	Benchmark	Baseline (%)	Ours (%)	Loss (pp)	Baseline (ms)	Ours (ms)	Speedup (×)
3B	MMStar	49.067	47.200	1.867	73.698	23.742	3.104
3B	AI2D	76.004	73.867	2.137	90.422	27.867	3.245
3B	MMBench-EN	71.769	71.684	0.085	65.794	22.429	2.933
3B	CV-Bench	72.354	68.044	4.310	112.385	29.552	3.803
3B	VisualPuzzles	31.079	30.565	0.514	89.813	28.422	3.160
3B	VisuLogic	24.200	25.700	-1.500	68.877	24.225	2.843
8B	MMStar	52.200	50.200	2.000	100.921	29.302	3.444
8B	AI2D	77.461	75.777	1.684	121.478	33.600	3.615
8B	MMBench-EN	75.595	74.150	1.446	90.951	28.308	3.213
8B	CV-Bench	74.869	70.678	4.191	151.967	35.613	4.267
8B	VisualPuzzles	29.794	29.538	0.257	121.053	34.504	3.508
8B	VisuLogic	23.800	25.700	-1.900	94.430	30.270	3.120
14B	MMStar	55.467	54.000	1.467	123.660	35.138	3.519
14B	AI2D	78.595	77.137	1.457	148.597	39.230	3.788
14B	MMBench-EN	77.041	75.255	1.786	110.754	33.981	3.259
14B	CV-Bench	74.118	69.669	4.448	188.824	41.011	4.604
14B	VisualPuzzles	33.305	31.764	1.541	148.558	40.230	3.693
14B	VisuLogic	25.700	26.500	-0.800	115.933	36.186	3.204

The frozen configuration uses 336-pixel inputs, full prompts and all layers, derived mixed precision, QKV fusion, and vision graph execution. Request-stream timing includes initial capture and request-time compilation; model loading and weight preparation are separate. Each request recomputes visual features. This does not isolate same-precision operator gains. MMBench-EN uses provided-pass group accuracy, not full official CircularEval; 53 groups are incomplete. CV-Bench uses source-balanced accuracy; the other benchmarks use forced-choice exact accuracy. Three repetitions are timing replicates, not independent training seeds.

Table 9. Gemma3 supplementary full-benchmark results. Loss is baseline accuracy minus method accuracy (percentage points).

Model	Benchmark	Baseline (%)	Ours (%)	Loss (pp)	Throughput (×)
4B	MMStar	46.000	46.000	0.000	3.086
4B	AI2D	70.499	70.499	0.000	3.410
4B	MMBench-EN	68.027	68.027	0.000	3.059
4B	CV-Bench	57.535	57.535	0.000	3.408
4B	VisualPuzzles	30.479	30.479	0.000	3.400
4B	VisuLogic	26.500	26.500	0.000	3.370
12B	MMStar	54.600	54.733	-0.133	2.428
12B	AI2D	79.922	79.922	0.000	2.698
12B	MMBench-EN	76.105	75.935	0.170	2.373
12B	CV-Bench	68.959	68.959	0.000	2.724
12B	VisualPuzzles	28.682	28.682	0.000	2.673
12B	VisuLogic	25.400	25.500	-0.100	2.601
27B	MMStar	56.867	56.867	0.000	1.885
27B	AI2D	81.218	81.250	-0.032	2.095
27B	MMBench-EN	80.952	80.952	0.000	1.857
27B	CV-Bench	71.434	71.434	0.000	2.127
27B	VisualPuzzles	33.904	33.904	0.000	2.050
27B	VisuLogic	24.900	24.900	0.000	2.007

Inputs remain at 896 pixels with BF16 and all layers. The method combines B2 graph execution, CPU input pipelining, and normalization-related fusion against a serial HF baseline. Ratios describe request-stream throughput, not single-request latency or matched-batch kernel gains. Only the complete 27B rerun is used; the earlier OOM cell is invalid. Equal accuracy does not mean identical predictions. MMBench-EN uses provided-pass group accuracy, not full official CircularEval; 53 groups are incomplete. CV-Bench uses source-balanced accuracy; the other benchmarks use forced-choice exact accuracy. Three repetitions are timing replicates, not independent training seeds.

Table 10. Bounded Ministral3-8B pilot studies and failure cases. Passes denote redundant precapture forward passes removed by the revised implementation.

Configuration	Evaluation set	Speedup (×)	Loss (pp)	Outcome
Mixed precision, 336	Independent set A (64)	3.087	3.125	Earlier confirmation
Mixed precision, 280	Reused screening set (64)	3.288	3.125	Screening only
Mixed precision, 224	Numeric fixture (32)	–	–	Invalid: agreement below 95%
Mixed precision, 280	Independent numeric set (32)	–	–	Invalid: agreement below 95%
Original precision, 224; 3 passes	Reused screening set (64)	1.996	4.6875	Capture overhead
Original precision, 280; 3 passes	Reused screening set (64)	2.396	0	Capture overhead
Original precision, 224; 0 passes	Reused screening set (64)	2.952	4.6875	Execution gates passed
Original precision, 280; 0 passes	Reused screening set (64)	2.963	0	Selected before confirmation
Original precision, 280; 0 passes	Independent set C (64)	2.952	4.6875	Confirmed; earlier setup retained

Invalid candidates have no reported speed. These pilot studies do not replace the frozen full-benchmark configuration. Independent confirmation sets differ, so their score differences do not establish statistical superiority. One error on 64 examples corresponds to 1.5625 percentage points.

Reporting scope and reproducibility.

All tables are generated from archived result summaries. The Qwen main cohort is kept separate from earlier standalone measurements and delivered plugin versions. Other families are supplementary transfer/adaptation studies; joint changes do not identify individual component effects. Historical reports, including invalid or superseded runs, are retained for traceability rather than counted as additional accepted results. No new GPU measurements were performed for this formatting revision. Per-table precision follows the available source records; missing latencies are not reconstructed from rounded ratios.

The accompanying source manifest records the input hashes. The underlying model weights, checkpoints, and per-example predictions remain in their existing execution storage. This document reports scoped forced-choice visual question answering results, not unrestricted text-generation performance or a universally interchangeable plugin.